Lecture 06 (B05): Deep Equilibrium Nets — The Central Idea

Heterogeneity, high-dimensional state spaces, the DEQN solution

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Outline

1. **Motivation**
   Why heterogeneous-agent models? The curse of dimensionality.
2. **From ANNs to Deep Equilibrium Nets**
   DNNs as function approximators, supervised vs. unsupervised loss, the DEQN training algorithm.
3. **Brock–Mirman Benchmark**
   Analytical test case, DEQN implementation.
4. **Parallelization & HPC**
   Data parallelism, MPI/Horovod, scalability to many nodes.
5. **Hands-on exercises**
   Notebooks 03/04 extend Brock–Mirman to uncertain, constrained, and multi-agent settings (endogenous labor, KKT multipliers with Fischer–Burmeister, 6-period OLG).

Wrap-up: **hands-on notebooks** on Nuvolos Cloud. Notebook 01 reproduces the slide derivation; 02 adds AR(1) productivity shocks; 03/04 are the blank/solved versions of the morning's exercises.

Roadmap of This Lecture (75 min)

Motivation (14 min)

- Heterogeneity and high dimensions
- Nonlinearities and kinks
- Irregular ergodic sets
- Our solution: DEQNs

From ANNs to DEQNs (25 min)

- DNNs as function approximators
- Supervised vs. unsupervised loss
- The DEQN training algorithm

Brock–Mirman Benchmark (18 min)

- Analytical test case
- DEQN implementation

Parallelization & HPC (10 min)

- Data parallelism (MPI/Horovod)
- Scalability to many nodes

Wrap-up + Hands-on (10 min)

- Four notebooks: BM (01), BM+shocks (02), exercises blank/solved (03/04)
- Exercise preview: KKT + Fischer–Burmeister, 6-period OLG

Part I

Motivation

Heterogeneity Matters

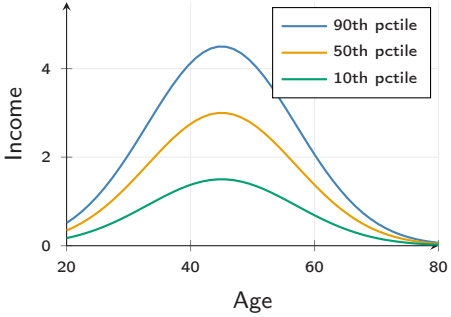
Key questions: inequality over the life cycle; idiosyncratic vs. aggregate risk; distributional effects of policy.

Quantitative punchline:

- Hand-to-mouth HH: MPC ≈ 1; unconstrained: MPC ≈ 0.05
- Representative agent averages these out, misses the policy response

Kaplan, Moll & Violante 2018; Bilbiie 2008; Krueger et al. 2016

$\Rightarrow$ Heterogeneity is one of several forces that blow up d (also: many countries, many assets, climate state, . . .).



Stylized income-by-age profiles

Dynamic Stochastic General Equilibrium Models

Generic formulation: find policy functions $p(\cdot)$ such that

$$\mathbb{E}\Big[G\big(\boldsymbol{x}_t,\, \boldsymbol{x}_{t+1},\, p(\boldsymbol{x}_t),\, p(\boldsymbol{x}_{t+1})\big) \;\Big|\; \boldsymbol{x}_t,\, p(\boldsymbol{x}_t)\Big] = 0$$

with transition law $\boldsymbol{x}_{t+1} = h\big(\boldsymbol{x}_t,\, p(\boldsymbol{x}_t),\, \varepsilon_{t+1}\big)$.

Four computational challenges (Azinovic, Gaegauf & Scheidegger, 2022):

1. Stochasticity: nontrivial expectations over ε_{t+1} and the ergodic distribution
2. High-dimensional state space: heterogeneity, many assets, many shocks all push d up
3. Strong nonlinearities / kinks from occasionally binding constraints and regime switches
4. Irregular geometry of the ergodic set: recurrent states are not a hypercube

No classical method addresses all four jointly: sparse grids cover (1-3) but fail past $d \sim 20$; Gaussian processes cover (1,2,4) but choke on large data.

What is High-Dimensional?

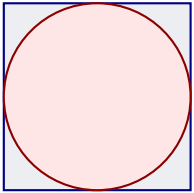
Dimensions d	Grid points n^d	Runtime
1	10	10 seconds
2	100	2 minutes
5	10^5	1 day
10	10^{10}	317 years
20	10^{20}	$\sim$ 3 trillion years

Assumes $n = 10$ grid points per dimension and 1 second per evaluation.

Three remedies:

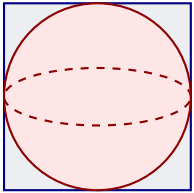
1. Sparse grids (Brumm & Scheidegger, 2017)
2. Random grids (Rust, 1997)
3. Neural networks (this lecture)

Volumes in High Dimensions



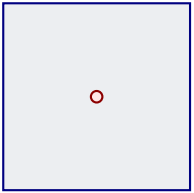
2D

$V_{\text{sphere}}/V_{\text{cube}} = \pi/4 \approx 0.785$



3D

$V_{\text{sphere}}/V_{\text{cube}} = \pi/6 \approx 0.524$



100D

$V_{\text{sphere}}/V_{\text{cube}} \approx 1.87 \times 10^{-70}$

Implication

In high dimensions, almost all volume is in the corners. Grid-based methods waste most of their evaluation points in regions that barely contribute to the integral or approximation.

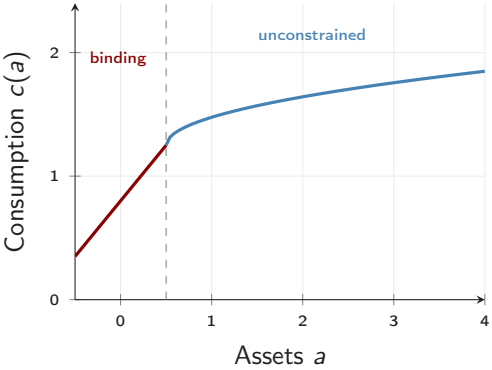
Nonlinearities and Kinks

Where do kinks come from?

- Borrowing constraints: $a_{t+1} \geq \underline{a}$
- Zero lower bound: $i_t \geq 0$
- Short-sale / participation: corner solutions
- Default, regime switches, indivisibilities

Occasionally binding constraints $\Rightarrow$ policy functions are non-smooth at the boundary of the binding region.

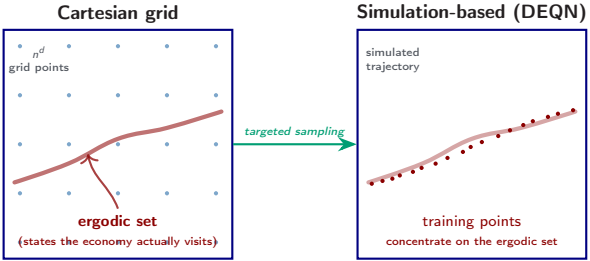
Polynomial / spectral methods lose accuracy; sparse grids can resolve kinks but only for $d \lesssim 20$. Neural networks handle kinks natively via ReLU-type activations.



Kinked consumption policy at the borrowing limit

The Ergodic Set Is Not a Hypercube

Grid-based methods tile the full hypercube $\boldsymbol{x} \in [\boldsymbol{x}_{\min}, \boldsymbol{x}_{\max}]^d$. **But** simulated states drawn from the model's law of motion fill only a thin, irregular subset, the ergodic set (states actually visited by the economy), often $\ll 1\%$ of the hypercube (Judd, Maliar & Maliar, 2011).



A full Cartesian grid wastes most of its work in regions the economy never visits $\Rightarrow$ motivates simulation-based, grid-free methods.

Abstract Problem Formulation

We need a numerical method that satisfies:

1. **Global approximation:** avoid purely local perturbation solutions
2. **Approximate solutions:** accept small residual error $\varepsilon > 0$
3. **Grid-free in practice:** avoid explicit tensor-product growth n^d
4. **Flexible geometry:** handle nonlinearities, kinks, and irregular ergodic sets
5. **Parallel-friendly:** benefit from parallelization on modern hardware

$\Rightarrow$ **DEQNs address this wishlist well in practice,** while mitigating rather than eliminating high-dimensional costs.

Our Solution: Deep Equilibrium Nets

Core idea (Azinovic, Gaegauf & Scheidegger, 2022):

Replace the traditional grid-based solution with a neural network trained to satisfy economic equilibrium conditions.

Three Key Ingredients

1. **Equilibrium error as loss function:** use the economic equations $G(\cdot) = 0$ directly, with no labeled data needed
2. **Stochastic gradient descent:** optimize network parameters ρ via SGD on mini-batches
3. **Simulated paths:** draw training points from the ergodic distribution of the model itself

$\Rightarrow$ This turns solving an economic model into training a neural network.

It avoids explicit state-space grids, but expectation accuracy, sampling quality, and optimization still matter.

Part II

From ANNs to Deep Equilibrium Nets